

ECE4010J Probabilistic Methods in Engineering (Summer 2026)

Assignment 8

Date Due: See canvas

This assignment has a total of **(125 points)**.

Note: Unless specified otherwise, you must show the details of your work via logical reasoning for each exercise. Simply writing a final result (whether correct or not) will receive **0 point**.

Exercise 8.1 (15 pts) In order to be effective, reflective highway signs must be picked up by the automobile's headlights. To do so at long distances requires that the beams be on "high". A study conducted by highway engineers reveals that 45 of 50 randomly selected cars in a high-traffic-volume area have the headlights on low beam.

- (i) (5 pts) Find a point estimate for p , the proportion of automobiles in this type area that use low beams.
- (ii) (5 pts) Find a 90% confidence interval on p .
- (iii) (5 pts) How large a sample is required to estimate p to within 0.02 with 90% confidence?

Exercise 8.2 (15 pts) A new computer network is being designed. The makers claim that it is compatible with more than 99% of the equipment already in use.

- (i) (5 pts) Set up the null and alternative hypotheses needed to get evidence to support this claim.
- (ii) (5 pts) A sample of 300 programs is run, and 298 of these run with no changes necessary. That is, they are compatible with the new network. Can H_0 be rejected? Explain, based on the p -value of the test.
- (iii) (5 pts) What practical conclusion can be drawn on the basis of your test?

Exercise 8.3 (15 pts) A random sample of 500 workers engaged in research and development (R & D) last year is selected. Of these, 178 earn over \$72,000 per year. Of the 450 workers in R & D studied during the current year, 220 earn in excess of \$72,000 per year.

- (i) (5 pts) Let p_1 and p_2 denote the proportion of workers engaged in research and development who earned over \$72,000 per year last year and this year, respectively. Find point estimates for p_1 , p_2 , and $p_1 - p_2$.
- (ii) (5 pts) Find a 95% confidence interval for $p_1 - p_2$.
- (iii) (5 pts) Would you be surprised to hear someone claim that the proportion of R & D workers earning over \$72,000 was the same this year as it was last year? Explain, on the basis of the confidence interval of part (ii).

Exercise 8.4 (15 pts) The cost of correcting a defect in a bipolar digital integrated circuit depends on when the defect is discovered. If it is discovered before it is integrated into a computer system, the cost may be only pennies. However, if it is not found until after the device is in the field it could cost thousands of dollars to repair. The electrical defect rate of two types of circuits produced by a particular company is being studied. It is suspected that the defect rate of their ALS circuits (advanced lower-power Schottky) is smaller than that of their LPS circuits (lower-power Schottky).

- (i) (5 pts) Let p_1 and p_2 denote the proportions of ALS circuits and LPC circuits produced, respectively, that have electrical defects. Set up the null and alternative hypotheses needed to confirm their suspicions.
- (ii) (5 pts) What is the critical point for an $\alpha = 0.1$ level test?
- (iii) (5 pts) Two thousand circuits of each type are randomly selected and tested. It is found that three of the ALS and five of the LPS circuits have electrical defects. Estimate p_1 , p_2 , and $p_1 - p_2$. Based on these data, can H_0 be rejected at the $\alpha = 0.1$ level?

Exercise 8.5 (25 pts) To decide whether or not to purchase a new hand-held laser scanner for use in inventorying stock, tests are conducted on the scanner currently in use and on the new scanner. These data are obtained on the number of 7-inch bar codes that can be scanned per second:

New	Old
$n_1 = 61$	$n_2 = 61$
$\bar{x}_1 = 40$	$\bar{x}_2 = 29$
$s_1^2 = 24.9$	$s_2^2 = 22.7$

- (i) (5 pts) Test $H_0 : \sigma_1^2 = \sigma_2^2$ at the $\alpha = 0.2$ level to be sure that pooling is appropriate.
- (ii) (5 pts) Find s_p^2 .

- (iii) (5 pts) Find a 90% confidence interval on $\mu_1 - \mu_2$.
- (iv) (5 pts) Does the new laser appear to read more bar codes per second on the average? Explain.
- (v) (5 pts) Since the number of bar codes that can be scanned per second is discrete, we have not satisfied the normality requirement. What theorem justifies the use of the pooled T procedure in this case?

Exercise 8.6 (15 pts) Strontium-90, a radioactive element produced by nuclear testing, is closely related to calcium. In dairy lands, strontium-90 can make its way into milk via the grasses eaten by dairy cows. It then becomes concentrated in the bones of those who drink the milk. In 1959 a study was conducted to compare the mean concentration of strontium-90 in the bones of children to that of adults. It was thought that the level in children was higher because the substance was present during their formative years.

Children	Adults
$n_1 = 121$	$n_2 = 61$
$\bar{x}_1 = 2.6$ picocuries per gram	$\bar{x}_2 = 0.4$ picocuries per gram
$s_1^2 = 1.44$	$s_2^2 = 0.0121$

- (i) (5 pts) Set up the null and alternative hypotheses needed to verify this contention.
- (ii) (5 pts) Based on these data, is pooling appropriate?
- (iii) (5 pts) Test the null hypothesis of part (i). Can H_0 be rejected? Explain, based on the p -value of your test. What practical conclusion can be drawn from these data?

Exercise 8.7 (15 pts) Information about ocean weather can be extracted from radar returns with the aid of a special algorithm. A study is conducted to estimate the difference in wind speed as measured on the ground and via the Seasat satellite. To do so, wind speeds are measured using the two methods simultaneously at 12 specified times. These data result:

Windspeed, m/s			
Time	Ground (x)	Satellite (y)	Differences $d = x - y$
1	4.46	4.08	
2	3.99	3.94	
3	3.73	5.00	
4	3.29	5.20	
5	4.82	3.92	
6	6.71	6.21	
7	4.61	5.95	
8	3.87	3.07	
9	3.17	4.76	
10	4.42	3.25	
11	3.76	4.89	
12	3.30	4.80	

- (i) (5 pts) Find the difference scores for the above data subtracting in the order indicated.
- (ii) (5 pts) Find $\bar{d}$ and s_d .
- (iii) (5 pts) Find a 95% confidence interval on the mean difference in measurements taken by these methods. Based on this interval, is there reason to believe that, on the average, the satellite measurements differ from those taken on the ground? Explain.

Exercise 8.8 (10 pts) A study is conducted to determine the effect of acid rain and other industrial pollutants on lake water. Random samples are drawn from 10 lakes in a heavily industrialized area and from eight lakes in a primitive forested area. These data are obtained on the pH of the water:

Industrial area (I)		Primitive area (P)	
6.9	7.0	7.0	6.8
6.2	6.5	6.9	7.1
6.3	6.6	6.7	7.0
5.9	5.5	7.1	7.2
6.0	7.3		

At the $\alpha = 0.025$ level, can we claim that the pH of the water in the industrialized area tends to be lower than that in the primitive area?